

CSCE 3530 – Introduction to Computer Networks

Final Project Proposal

Student Name: Genevieve

Course: CSCE 3530 – Introduction to Computer Networks

Instructor: Dr. Zarafshani

Submission: Project Proposal

Project Title

Design and Simulation of a Virtual Enterprise Network Using CISCO packet tracker and Wireshark

Project Overview

The purpose of this project is to design, simulate, configure, and analyze a virtual computer network environment using GNS3 and Wireshark. The project will provide practical experience in understanding how network devices communicate and how networking protocols function together in a real-world environment.

This project will involve creating a virtual enterprise network containing routers, switches, client computers, and servers. The network will be configured to demonstrate important networking protocols including DHCP, DNS, ARP, HTTP, HTTPS, ICMP, TCP, UDP, NAT, SSL/TLS, and IP addressing.

The project will also use Wireshark packet analysis to capture network traffic and observe communication between devices. This project will help demonstrate how modern computer networks are designed, tested, and managed.

Project Objectives

The objectives of this project are to:

- Build a virtual network topology using GNS3
 - Configure routers, switches, servers, and client computers
 - Implement automatic IP address assignment using DHCP
 - Configure DNS services for hostname resolution
 - Configure HTTP and HTTPS web communication
 - Demonstrate communication using TCP and UDP protocols
 - Test network connectivity using ICMP commands such as ping and traceroute
 - Capture and analyze network traffic using Wireshark
 - Demonstrate ARP communication and MAC address resolution
 - Implement NAT for communication between internal and external networks
 - Document network communication and protocol interactions
-

Proposed Network Design

The proposed network will simulate a small enterprise network environment. The network will contain multiple client computers, switches, routers, and servers connected together to allow communication between internal devices and external internet access.

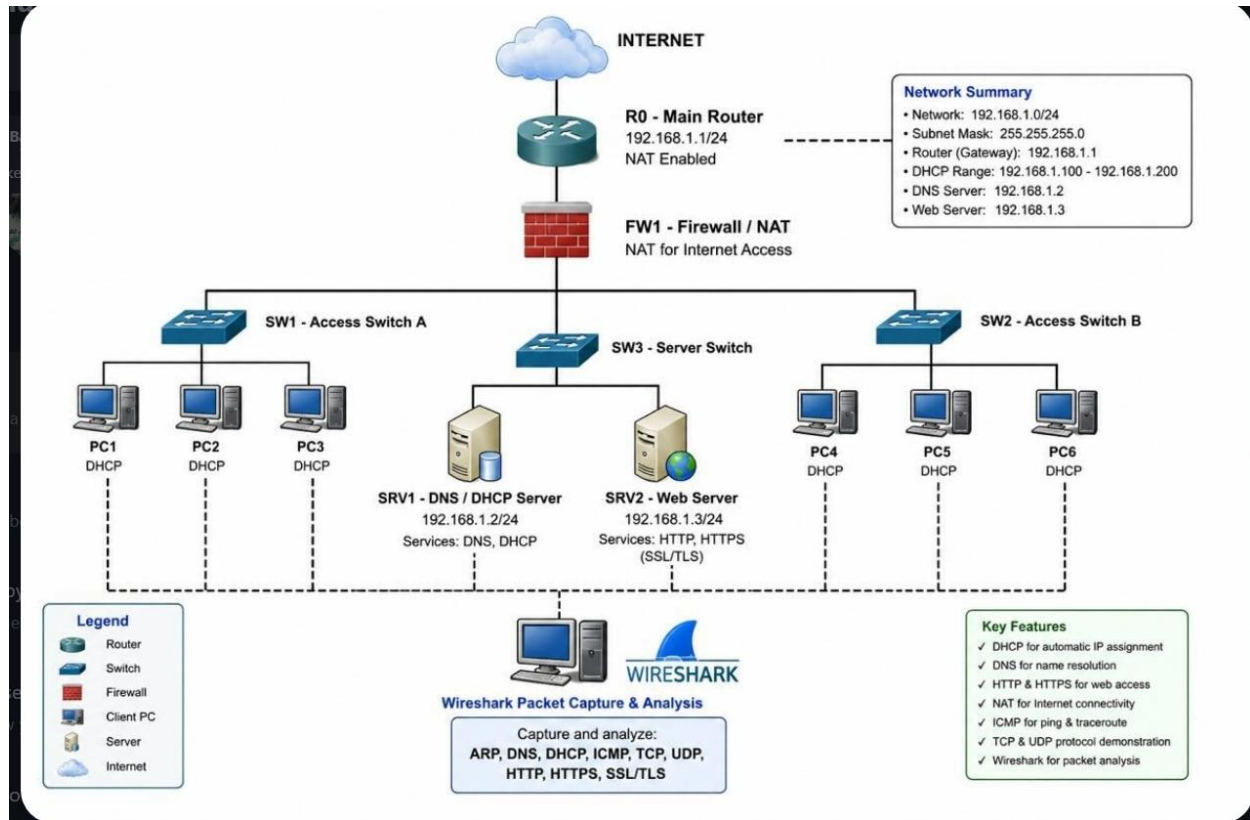
The network will demonstrate routing, switching, DNS resolution, DHCP automatic IP assignment, web server communication, secure HTTPS communication, packet capture analysis, and NAT translation.

The network will contain:

- One Main Router
- Two Network Switches
- Six Client Computers
- One DNS/DHCP Server
- One Web Server

- Internet Cloud Simulation
- Firewall/NAT Routing

Network Topology Diagram



IP Addressing Plan

The project will use private IPv4 addressing.

Device	IP Address
Main Router	192.168.1.1
DNS Server	192.168.1.2
Web Server	192.168.1.3
Client PCs	DHCP Assigned

Device	IP Address
Subnet Mask	255.255.255.0

Network Address:

192.168.1.0/24

Protocols To Be Demonstrated

The following protocols will be implemented and demonstrated:

DHCP

DHCP will automatically assign IP addresses to client computers.

DNS

DNS will resolve hostnames into IP addresses for communication.

ARP

ARP will identify MAC addresses for local network communication.

HTTP

HTTP will allow communication between clients and the web server.

HTTPS

HTTPS will demonstrate encrypted secure communication using SSL/TLS.

ICMP

Ping and traceroute commands will test network connectivity.

TCP

TCP communication will be demonstrated using reliable data transmission during web communication.

UDP

UDP communication will be demonstrated through DNS communication.

NAT

NAT will translate internal private addresses for external communication.

SSL/TLS

SSL/TLS will provide encryption for secure HTTPS traffic.

Wireshark Packet Analysis Plan

Wireshark will be used to monitor and capture network traffic generated during testing.

The packet analysis will include:

- DHCP request and response packets
- DNS query and response packets
- ARP request and reply packets
- ICMP ping packets
- TCP connection establishment (Three-Way Handshake)
- UDP communication packets
- HTTP request and response packets
- HTTPS encrypted communication packets

Packet analysis will help identify how devices communicate and how protocols operate during data transmission.

Testing Procedures

The network will be tested using the following methods:

- Ping testing between all devices
- Traceroute testing
- Browser testing using HTTP and HTTPS
- DNS hostname testing
- Packet capture using Wireshark
- Connectivity verification between internal and external networks

- Routing and NAT functionality testing
-

Expected Outcomes

At the completion of this project, the network should successfully:

- Automatically assign IP addresses using DHCP
 - Resolve hostnames using DNS
 - Demonstrate successful client-server communication
 - Demonstrate HTTP and HTTPS communication
 - Show TCP and UDP communication between devices
 - Capture and analyze packets using Wireshark
 - Demonstrate NAT and routing functionality
 - Verify connectivity using ICMP ping and traceroute
-

Conclusion

This project will provide practical experience in network design, network configuration, protocol implementation, troubleshooting, and packet analysis. Through the use of GNS3 and Wireshark, the project will demonstrate how modern computer networks operate and how networking devices communicate using essential networking protocols.

The project will strengthen understanding of real-world network infrastructure while providing hands-on experience with industry-standard networking tools and technologies.